

AI & Machine Learning

BRANCH: COMPUTER SCIENCE (CSE)

COURSE CODE: CS401

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for AI & Machine Learning (CS401) under the Computer Science (CSE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Introduction & Supervised Learning

Machine Learning vs traditional programming. Linear Regression, Logistic Regression. Classification: k-Nearest Neighbors (k-NN), Decision Trees, Support Vector Machines (SVM), Naive Bayes.

Unit 2: Unsupervised Learning

Clustering: K-Means, Hierarchical Clustering, DBSCAN. Dimensionality Reduction: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA).

Unit 3: Artificial Neural Networks

Perceptron, Multi-Layer Perceptron (MLP). Backpropagation algorithm. Activation functions (Sigmoid, ReLU, Tanh). Intro to Deep Learning: CNNs, RNNs.

Unit 4: Evaluation & Optimization

Bias-Variance Tradeoff. Overfitting and Underfitting. Evaluation metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC. Regularization: L1 (Lasso) and L2 (Ridge).

*Mean Squared Error (MSE), Sigmoid activation: $1 / (1 + e^{-x})$, F1-Score = $2 * (Prec * Rec) / (Prec + Rec)$.*

CRITICAL FORMULA / PRINCIPLE:

Key Formulas & Exam Pointers



